DDD-T-531H
October 14, 1988
SUPERSEDING
DDD-T-531G
February 2, 1978

FEDERAL SPECIFICATION

TOWEL, HAND; AND CLOTH, HUCK, COTTON

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand towels and cotton huck cloth toweling.

1.2 Classification.

1.2.1 Types, classes, sizes, and colors. Towels and toweling shall be the following types, classes, sizes and colors as specified (see 6.2):

Type I - Towel, hand

Class 1 - With woven design and stripe

Color - Green

Class 2 - Without woven design and stripe

Color - White
Color - Green

Class 3 - With stamped U.S., without stripe

Color - White
Color - Green

Size 1 - 17 inches by 36 inches
Size 2 - 17 inches by 17 inches

Type II - Cloth, toweling, huck, cotton

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7210

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

V-T-276	-	Thread, Cotton
V-T-295	-	Thread, Nylon
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-P-1134	-	Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Federal Standards:

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specifications:

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MS 16686	-	Medical Services Symbol

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FEDERAL REGULATIONS

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20402-0001. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished dyed towels and toweling shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials (see 6.6).

3.3.1 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn, and spun into either singles or 2 ply as specified in table I.

3.4 Physical requirements. All towels and toweling shall conform to the requirements shown in table I when tested as specified in 4.4. Unless otherwise specified by civil agencies (see 6.2), towels shall be of the size specified in table I.

TABLE I. Physical requirements

Type	Size	Dimensions (minimum) Width 1/	Length	Weight minimum	Yarns per inch, minimum Warp	Filling	Yarn ply Warp	Filling	Breaking strength (lbs) minimum Warp	Filling
I	size 1	17 in	36 in	2.55 lb/doz	52	26 (as singles) or 52 (in pairs)	1 3/ or 2	1 or 2	70	60
	size 2	17 in	17 in	1.2 lb/ doz	52	26 (as singles) or 52 (in pairs)	1 3/ or 2	1 or 2	70	60
II		17 in	(see 3.12)	7.0 oz/ sq yd	52	26 (as singles) or 52 (in pairs)	1 or 2	1 or 2	70	60

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1/ The width shall be inclusive of selvages. The selvages shall be 1/8 inch to 1/4 inch wide.

2/ Sizes other than 17 inches by 36 inches shall have a proportionate weight.

3/ Except that the warp yarns for type I, class 1 towels shall be 2 ply.

3.5 Weave. The weave for type I towels and type II toweling shall be a plain huck weave. Type I, classes 1, 2, and 3 towels shall have a border at each end of the towel. The border shall be either a fancy woven pattern 1 ± 1/8 inch wide located 1 1/2 inches ± 1/8 inch above the hem, or a plain woven border 2 inches ± 1/4 inch deep including hem. The type I, class 3, color green towels shall be continuous plain huck weave without a border. The class 1 towel shall have a woven stripe with design as specified in 3.6.

3.6 Design of stripe for type I, class 1 towels. The class 1 towels shall have a Green Gray 180 stripe 1 1/2 inches ± 1/8 inch wide, woven with a 5-harness satin weave lengthwise through the center of the towel. Centered in the green stripe there shall be a woven filling effect design, consisting of a caduceus and letters "U" and "S," conforming to MS 16686. The caduceus shall be 1 1/4 inches ± 1/16 inch high and 1 1/4 inches ± 1/16 inch wide. Block letters "U" and "S" shall be 1/4 inch ± 1/16 inch from the staff of the caduceus. Each letter shall be 9/16 inch ± 1/16 inch high and 7/16 inch ± 1/16 inch wide. The design shall be woven with the letters "U" and "S" and the staff of the caduceus at right angles to the length of the towel.

3.7 Color.

3.7.1 *Towels and toweling*.

3.7.1.1 Type I, class 1, color-green. The body and the stripe of the class 1 towel shall be Green Gray 180.

3.7.1.2 Type I, class 2, color-white. Class 2, color-white towels shall be bleached white.

3.7.1.3 Type I, class 2, color-green. Class 2, color-green towels shall be dyed Green Gray 180.

3.7.1.4 Type I, class 3, color-white. Class 3, color-white towels shall be bleached white.

3.7.1.5 Type I class 3, color-green. Class 3, color green towels shall be dyed Green Gray 180.

3.7.1.6 Type II. Type II toweling shall be bleached white or dyed as specified (see 6.2).

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3.7.2 Sulfur content. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished towels or cloth shall contain no more labile sulfur than that shown by the standard sample when tested as specified in 4.4.4. When a standard sample is not available, the dyed and finished towels or cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.4.

3.7.3 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.7.4 Colorfastness. The dyed towels and toweling shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample (see 6.3). When no standard sample is available, the dyed towels and toweling shall show "good" fastness to laundering (after 3 cycles) when tested as specified in 4.4.4.

3.8 Nonfibrous material. The finished towels and cloth toweling shall contain not more than 5.0 percent starch and protein including chloroform-soluble and water-soluble material when tested as specified in 4.4.4.

3.9 Construction.

3.9.1 Hems and selvages. The finished towel shall be hemmed at each end. The sides of the towel or toweling shall have two fast selvages, one fast selvage and one hemmed side, or two hemmed sides. There shall be no raw edges showing on the finished towel.

3.9.2 End hems. The hem at each end of the towel shall be 1/4 inch to 3/8 inch wide. Stitching shall be 1/16 inch to 1/8 inch from the turned edges, 10 to 12 stitches per inch using stitch type 301 or 7 to 9 stitches per inch using stitch type 401, and seam type EFb-1 of FED-STD-751. At the beginning and at the end of each hem, the hem with the 301 stitch shall be backstitched at least 1/2 inch.

3.9.3 Side hems. When side hems are used, they shall be 1/4 inch to 3/8 inch wide. Stitching shall be 1/16 inch to 1/8 inch from the turned edges, 10 to 12 stitches per inch using stitch type 301 or 7 to 9 stitches per inch using stitch type 401 and seam type EFb-1 of FED-STD-751.

3.10 Thread. The cotton thread for hemming shall be bleached white or dyed to match the shade of the towel (see 3.7) and shall conform to type IA2 or IA3 of V-T-276. The thread used with stitch type 301 shall be ticket No. 50, 3 ply. The thread used with stitch type 401 shall be ticket No. 50, 3 ply for the needle and ticket No. 70, 2 ply for the looper. The dyed thread shall show "good" fastness to laundering (after 3 cylces).

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3.10.1 Alternate nylon thread for the 301 stitch and 401 stitch. A textured polyester thread conforming to type II, class 1, size 80 of MIL-T-44193 may be used as an alternate thread for the 301 and the 401 stitch. The thread shall be white or dyed to match the shade of the towel. The dyed thread shall show good colorfastness to laundering. All testing shall be as specified in V-T-295, except that a 1.0 ounce pretension load shall be applied before closing the clamp in the breaking strength machine.

3.11 Marking.

3.11.1 Type I, class 2 towels. When identification marking is specified (see 6.2), each class 2 towel shall be marked on one corner with a marking conforming to type IV, class 5 of DDD-L-20.

3.11.2 Type I, class 3 towel. Each class 3 towel shall be permanently printed or stamped with 3/4 (+ 1/8, -0) inch high block letters "U.S." in black color on one of the corners of the towel. The marking shall conform to type IV, class 9 of DDD-L-20. The marking shall show fastness to accelerated laundering.

3.12 Length and put-up for type II towels.

3.12.1 Civil agencies (bolt put-up). Fifty ± 2.5 yards of cloth toweling shall be put-up in bolt form or, alternately at the option of the manufacturer, the length of toweling may be put-up by winding on a flat chipboard or cardboard insert of not less than 1/8 inch thick by 12 inches wide. Toweling 46 inches or less in width shall be folded open width in 1-yard folds with the face inward. When the specified number of yards for the bolt have been folded, the entire bolt shall be folded in half with crease at right angles to the selvages. Toweling over 46 inches up to 60 inches in width inclusive shall be folded in half lengthwise with face inward, selvage to selvage, then flat-folded in 1-yard folds. The entire bolt so folded shall then be folded in half with crease at right angles to the selvages. Toweling over 60 inches in width shall be folded in quarters lengthwise, selvages together with face inward, then flat folded in 1-yard folds. The entire bolt so folded shall then be folded in half with crease at right angles to the selvages.

3.12.2 Military requirements. Unless otherwise specified (see 6.2), the type II toweling shall be in one continuous piece of not less than 40 yards and shall be put-up in rolls as specified in PPP-P-1134.

3.13 Fiber identification. Each type I towel and each roll or bolt of type II toweling shall be labeled for fiber content in accordance with the Textile Fiber Products Identification Act.

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3.14 Workmanship. The finished towels and cloth toweling shall conform to the quality established by this specification. The occurrence of defects for type I towels shall not exceed the applicable acceptance quality levels specified. For type II toweling, the points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.2 End item examination (type I).

4.4.2.1 End item screening examination. The production of towels shall be visually screened by the contractor and sorted into lots designated as first and second quality (see 6.5) in accordance with standard industrial practice.

4.4.2.2 End item examination for acceptance of first quality lots. Lots designated by the contractor as first quality (see 4.4.2.1) shall be examined for acceptance by the Government. The lot size shall be expressed in units of one towel each. The sample unit shall be one completely fabricated towel. The inspection lot shall consist of all towels presented for inspection at one time.

4.4.2.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of towels. The sample unit shall be one towel. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defectives per hundred units, shall be 4.0. Any towel that is found to contain either or both of the following shall be classified as a defective unit: Any major defect listed in table II. More than two minor defects listed in table II.

TABLE II. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Quality of material	Cut, hole, or tear	X	
	Smash	X	
	Broken or missing yarn:		
	- two or more contiguous	X	
	- single, 4 inches or more in length		X
	Fine filling bar or light place	X	
	Hitchback:		
	- open or widest point four or more filling yarn diameters, or sequence of openings resulting in weak spot	X	
	- opening at widest point less than four filling yarn diameters or raised places		X

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TABLE II. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Quality of material - continued	Coarse filling bar or heavy place		X
	Slough-off, jerked-in filling, slub, slug, or gout:		
	- more than 1/8 inch in thickness		X
	- two or more per towel 1/16 inch to 1/8 inch thickness		X
	Floats or skips:		
	- multiple, 1/2 inch or more in combined directions	X	
	- multiple, less than 1/2 inch in combined directions or single floating 2 inches or more		X
	- single, 4 or more inches or more in length	X	
	Trash, mote, seed coat, or leaf fragments		X
	Spot, stain, or streak more than 1/2 square inch in area	X	
	Accumulation of small spots or stains (five or more per towel)		X
Workmanship	Selvage:		
	- not within width tolerance		X
	- cut, torn, or broken	X	
	- slack, tight, or scalloped		X
	Border missing at either end (when applicable)	X	
	Fancy border, when applicable, more than 1 1/8 inches or less than 7/8 inch width		X
	Fancy border pattern, when applicable, placed more than 1 5/8 inches or less than 1 3/8 inches above the hem		X
	Plain border, when applicable, including hem, more than 2 1/4 inches or less than 1 3/4 inches wide		X
	Not bleached, when bleach is specified		X
	Wrong color or off shade (body of towel or design)	X	

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TABLE II. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Workmanship - continued	Stripe (class 1):		
	- missing	X	
	- less than 1 3/8 inches or more than 1 5/8 inches wide		X
	Design (class 1):		
	- lettering or caduceus missing or incorrect	X	
	- caduceus and letters spaced more than 1/2 inch off center		X
	Marking (class 3 towel) less than 3/4 inch or more than 7/8 inch high		X
	Objectionable odor		X
	Seams and stitching:		
	- hem less than 1/4 inch or more than 3/8 inch wide		X
	- hem omitted	X	
	- raw edge showing		X
	- less than 10 or more than 12 stitches per inch		X
	- less than 7 or more than 9 stitches per inch for stitch type 401 on end hems		X
	- stitching less than 1/16 inch or more than 1/8 inch from turned edge		X
	Open seam:		
	- over 1 inch in length	X	
	- over 1/4 inch but less than 1 inch in length		X
	Wrong stitch or stitching type		X
	Loose stitch tension resulting in loosely secured seam		X
	Tight stitch tension (stitches break when normal strain is applied to seam or stitching)		X
	Backstitching at beginning and end of hem omitted or less than 1/2 inch		X
	Needle chew		X

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TABLE II. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Marking (when required)	Omitted, incomplete, illegible, incorrect, or misplaced		X

4.4.2.4 End item dimensional examination. The end items shall be examined for conformance with dimensions specified in table I. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of towels. The sample unit shall be one towel. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.2.5 Examination for acceptance of second quality lots when applicable (see 6.5). Lot designated by the contractor as second quality (4.4.2.1) shall be examined for acceptance by the Government. The sample unit shall be one completely fabricated towel. The inspection lot shall consist of all towels presented for inspection at one time.

4.4.2.5.1 Visual examination of second quality lots. The lot of towels shall be sampled in accordance with inspection level I of MIL-STD-105 and visually inspected. If any two towels in the sample are found to contain either or both of the following, the lot shall be rejected:

More than three of the major defects listed in table II.

More than seven total defects (majors and minor combined).

4.4.2.5.2 Dimensional examination of second quality lots. The lot of towels shall be examined for conformance with dimensional requirements in accordance with the procedures of 4.4.2.4.

4.4.3 Examination of the type II cloth toweling. Defects found during this examination shall be classified in accordance with 4.4.3.1 through 4.4.3.3.

4.4.3.1 Yard-by-yard examination. The entire yardage of each roll or bolt in the sample, but not more than 100 yards per roll or bolt, shall be examined on one side. Defects as defined in section I of FED-STD-4 which are clearly noticeable at a normal inspection distance of 3 feet shall be scored and assigned demerit points as listed in 4.4.3.1.1 except that only those slubs

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and knots which exceed the maximum limit shown in figure 1 of FED-STD-4 shall be scored. No individual linear yard from any one roll or bolt within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls or bolts selected from 20 containers. For determination of acceptability of the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceed the following:

35.0 points for dyed toweling
40.0 points for white toweling

For determination of acceptability on the individual roll or bolt quality basis, the lot shall be unacceptable if two or more individual rolls or bolts exceed the following:

53.0 points for dyed toweling
60.0 points for white toweling

If no individual roll or bolt exceeds the point level, the lot shall be accepted. If one roll or bolt exceeds the point level, a second sample of 20 rolls or bolts shall be selected for individual roll or bolt quality examination only. If no roll or bolt in the second sample exceeds the point value, the lot shall be accepted. If one or more rolls exceed the value, the lot shall be rejected. For individual roll or bolt quality examination, local type defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirement. Point computation for lot quality and individual roll or bolt quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in dimension	- four points

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The following defects, when present, shall be scored four points for each yard in which they occur:

Off shade or wrong shade, when color is specified.
Not bleached, when bleached is specified.
Uneven weaving throughout the roll or bolt, clearly noticeable.
Overall uncleanness.
Objectionable odor.
Width less than specified.
Selvage not within width tolerance.

4.4.3.2 Examination for length, type II only.

4.4.3.2.1 Civil agencies (individual bolts). Each bolt in the sample shall be examined for length. Any length found to be less than or more than the specified length including tolerances shall be considered as a defect with respect to length. The number of bolts taken in sampling (sample size) for this examination and the number of defects acceptable shall be in accordance with table III. The sample unit for this examination shall be one bolt.

TABLE III. Sample size and acceptance criteria

Lot size in yards	Sample size in rolls or bolts	Acceptance number
Up to 1,300 1/	3	0
1,301 up to and including 3,200	5	0
3,201 up to and including 8,000	7	0
8,001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	0
110,001	25	1

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.4.3.2.2 Military requirements (individual rolls). The roll shall be examined for length. Any length found to be less than the specified minimum length, or any length found to be more than 2 yards less than the length marked on the roll ticket or any roll not a continuous length shall be considered as a defect with respect to length. The sample unit for this examination shall be one roll. The number of rolls selected as sample (sample size) for this examination and the number of defects acceptable shall be in accordance with table III.

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4.4.3.2.3 Total yardage in sample - type II only. The lot shall be unacceptable if the total of the actual lengths of rolls or bolts in the sample is less than the total of the lengths marked on the roll or bolt tickets. The rolls (or pieces) examined shall be those selected for examination of individual rolls (or pieces).

4.4.3.3 Examination for compliance with Rules and Regulations Under the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll or bolt in the sample shall be examined for compliance with Rules and Regulations Under the Textile Fiber Products Identification Act. Each roll or bolt not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.4.4 Component testing. The end item shall be tested for the characteristics listed in table IV. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of 1 yard. The sample unit shall be 2 continuous yards full width of the finished cloth. The sample size (number of sample units) shall be as shown below. The lot shall be unacceptable if one or more sample units fail to meet any test requirement.

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Component tests

Characteristic	Requirement paragraph	Test method	Number of determinations	Results reported
Weight:				
Type I, lb/doz	3.4	Commercial	1	To the nearest 0.01 lb
Type II oz/sq yd	3.4	5041	-	-

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TABLE IV. Component tests - Continued

Characteristic	Requirement paragraph	Test method	Number of determinations	Results reported
Yarns per inch:				
Warp	3.4	5050	-	-
Filling	3.4	5050	-	-
Yarn ply:				
Warp	3.4	Visual	1	Pass or fail
Filling	3.4	Visual	1	Pass or fail
Breaking strength:				
Warp	3.4	5100	-	-
Filling	3.4	5100	-	-
Weave	3.5	Visual	1	Pass or fail
Weave in stripe	3.6	Visual	1	Pass or fail
Labile sulfur	3.7.2	2020	1	Pass or fail
Colorfastness to laundering (after 3 cycles)	3.7.4	5610 1/	-	-
Nonfibrous material	3.8	2611	-	-

1/ The specimens shall be dried after each of the 3 laundering cycles. Only the stain on the cotton fiber of the color transfer cloth shall be evaluated.

4.4.5 Packaging examination.

4.4.5.1 Type I. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container Polyethylene bag incorrectly fabricated, missing vent hole, or having open and noncontinuous heat-sealed seam and closure
Content	Number per container is more or less than required Number of towels per polyethylene bag is more or less than specified 1/ Different classes of towels in same polyethylene bag 1/

1/ For this defect three polyethylene bag shall be examined.

4.4.5.2 Type II. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that the preservation, packing, and marking requirements for type II toweling comply with the section 5 requirements.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Towels. Twelve or twenty four towels of one class and color only, folded in pairs with every six reversed, shall be neatly stacked in a bundle tied at each end with tape (see 6.2). Each bundle shall be placed in a snug-fitting flat (or square) style clear polyethylene film bag of 0.0015 inch thickness (± 20 percent tolerance). The bag shall be fabricated from polyethylene film tubing or sheeting. The polyethylene bag shall be formed with heat seal seams that are straight, continuous, and parallel to each other and to the formed edges of the bag. The bag closure shall be effected by heat sealing with the heat seal as close as possible to the open end. A 1/4-inch diameter hole shall be made at one corner of each bag to allow excess air to escape. Alternatively, closure may be effected by twisting and tying the open end of the bag with a metal band, plastic tie, or cord.

5.1.1.2 Toweling cloth. Toweling put up on rolls as specified in 3.12.2 shall be wrapped in accordance with the applicable level A requirements of PPP-P-1134.

5.1.2 Commercial. The towels and toweling shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Towels. Two hundred and fifty-two towels or when specified two hundred and eighty-eight towels of one class and color only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of the box shall be as specified in table V. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.1. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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TABLE V. Inside box dimensions

Size of towel	No. per box	Dimensions (± 1/4 inch)		
		Length	Width	Depth
17 x 17	288	27	15	9-1/2
17 x 36	252	27	15	17
17 x 36	288	27	15	19

5.2.1.2 Toweling cloth. Toweling shall be packed in accordance with the applicable level A requirements of PPP-P-1134.

5.2.2 Level B packing.

5.2.2.1 Towels. Two hundred and fifty-two towels, or when specified two hundred and eighty-eight towels of one class and color only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of the box shall be as specified in table V or 24-1/2 inches in length, 13-1/2 inches in width, and 18-1/2 inches in depth, respectively. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.5.1.

5.2.2.1.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.5.1.

5.2.2.2 Toweling cloth. Toweling shall be packed in accordance with the applicable level B requirements of PPP-P-1134.

5.2.3 Commercial packing. The towels and toweling, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), towels, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 6 in accordance with the appendix of MIL-STD-147.

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5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Military requirements.

5.4.1.1 Towels. In addition to any special marking required by the contract or purchase order, interior pack and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1.1.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

5.4.1.2 Toweling cloth. In addition to any special marking required by the contract or purchase order, interior packs, and shipping containers shall be marked in accordance with the applicable requirements of PPP-B-1134.

5.4.2 Civil agencies. In addition to any special marking required by the contract or purchase order, interior packs, and shipping containers shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The towels and toweling are intended for use by civil and military agencies of the Government.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, class, size and color required (see 1.2.1).
c. When a first article sample is required (see 3.2, 4.3, (and 6.4).
d. Size required (see 3.4).
e. Whether toweling is to be bleached or dyed (see 3.7.1.6).
f. Whether marking for type I, class 2 is required (see 3.11.1).
g. Length when other than specified (see 3.12.2).
h. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
i. Number of towels per bundle (see 5.1.1.1).
j. When 288 towels are required (see 5.2.1.1 and 5.2.2.1).
k. When palletization is required (see 5.3).
l. Type and class of unit load required (see 5.2.1.1).
m. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1.1).
n. Type marking required (see 5.4).

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6.3 Sample. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Distribution of second quality towels. Quantities and distribution of accepted second quality towels shall be in accordance with the terms of the contract or purchase order.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

6.7 Subject term (key word) listing.

Cloth, huck
Towels, hand
Toweling, cotton

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review Activities:

Army - MD
Navy - MC
Air Force - 84, 82
DLA - CT, DM

Civil Agency Coordinating Activity:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project No. 7210-0279

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DDD-T-531H	Towel, Hand; and Cloth, Huck, Cotton

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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